Zhiming Huang

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EMPLOYMENT

University of Washington, WA, USA — Dec 2025 - Present

- **Postdoctoral Scholar**, Paul G. Allen School of Computer Science & Engineering
- Advisor: Prof. Jamie Morgenstern

EDUCATION

University of Victoria, BC, Canada — Sep 2018 - Aug 2025

- **Ph.D.** in Computer Science (Aug 2020 - Aug 2025)
- Advisor: Prof. Jianping Pan
- Dissertation: Swap-Regret-Minimizing Bandits for Distributed Network Optimization
- **M.Sc.** in Computer Science (Sep 2018 - Aug 2020)
- Advisor: Prof. Jianping Pan
- Thesis: Thompson sampling-based online decision making in network routing

Northwestern Polytechnical University, Shaanxi, China — Sep 2014 - Jun 2018

- **B.Eng.** in Communication Engineering

RESEARCH INTERESTS

Online learning, game-theoretic learning, and trustworthy AI, with a focus on sequential decision-making and reliable prediction under limited feedback.

PUBLICATIONS

(α-β) denotes alphabetical author order.

PREPRINTS

[1] **Z. Huang**, J. Morgenstern, A. Roth and C. J. Zhang (α-β), Instance-Adaptive Online Multiclass Multicalibration, *arXiv:2610.15954*, Oct 2026. Preprint.

CONFERENCE PUBLICATIONS

[1] **Z. Huang**, J. Morgenstern, A. Roth and C. J. Zhang (α-β), Instance-Adaptive Online Multicalibration, *Advances in Neural Information Processing Systems (NeurIPS 2026)*, accepted.

[2] **Z. Huang**, B. Hu and J. Pan, Bridging the Regret Gap in Combinatorial Thompson Sampling: Worst-Case Guarantees and Algorithmic Refinement, *IEEE International Conference on Computer Communications (INFOCOM)*, May 2026.

[3] Q. Zhang, **Z. Huang**, J. Zhao, A. Ahangarpour and J. Pan, Faster Exploration and Exploitation for Communication Environment Awareness in Starlink, *IEEE International Conference on Computer Communications (INFOCOM)*, May 2026.

[4] **Z. Huang** and J. Pan, Faster Convergence for Unknown-game Bandits, *IEEE International Conference on Computer Communications (INFOCOM)*, May 2025.

[5] **Z. Huang** and J. Pan, Adversarial Semi-Bandits with Moving Arms, *IEEE International Conference on Computer Communications (INFOCOM)*, May 2025.

[6] B. Hu, **Z. Huang**, T. H. Zhang, M. Lécuyer and N. Hegde, Connecting Thompson Sampling and UCB: Towards More Efficient Trade-offs Between Privacy and Regret, *International Conference on Machine Learning (ICML)*, Jul 2025.

[7] Q. Zhang, **Z. Huang**, J. Zhao, and J. Pan, A Congestion Control Test Suite for Real-Time Communication, *ACM Multimedia Systems Conference (MMSys)*, Apr 2025.

[8] **Z. Huang**, B. Hu, and J. Pan, Gaussian Randomized Exploration for Semi-bandits with Sleeping Arms, *NeurIPS 2024 Workshop on Bayesian Decision-making and Uncertainty*, Dec 2024.

[9] F. Amri, **Z. Huang**, K. Liu, and J. Pan, Energy-aware Inter-data Center VM Migration over Elastic Optical Networks, *IEEE International Conference on Communication (Globecom)*, Dec 2023.

[10] **Z. Huang** and J. Pan, A Near-optimal High-probability Swap-Regret Upper Bound for Multi-agent Bandits in Unknown General-sum Games, *Uncertainty of Artificial Intelligence (UAI)*, Aug 2023.

[11] **Z. Huang**, K. Liu and J. Pan, End-to-end Congestion Control as Learning for Unknown Games, *IEEE International Conference on Distributed Computing Systems (ICDCS)*, Jul 2023.

[12] **Z. Huang**, B. Hu, and J. Pan, Poster: Multi-agent Combinatorial Bandits with Moving Arms, *IEEE International Conference on Distributed Computing Systems (ICDCS)*, Jul 2021.

[13] B. Hu, Y. Chen, **Z. Huang**, N. Mehta, and J. Pan, Intelligent Caching Algorithms in Heterogeneous Wireless Networks with Uncertainty, *IEEE International Conference on Distributed Computing Systems (ICDCS)*, Jul 2019.

[14] R. Liu, J. Song, **Z. Huang**, and J. Pan, EQRC: An Enhanced QR Code-based Secure E-coupon Transaction Framework, *IEEE International Conference on Communication (ICC)*, May 2019.

JOURNAL PUBLICATIONS

[1] W. Yang, L. Cai, S. Shu, A. Sepahi, **Z. Huang** and J. Pan, QoS-driven Contextual MAB for MPQUIC Supporting Video Streaming in Mobile Networks, *IEEE Transactions on Mobile Computing (TMC)*, Nov 2024.

[2] **Z. Huang** and J. Pan, Game-theoretic Bandits for Network Optimization with High-probability Swap-regret Upper Bounds, *IEEE/ACM Transactions on Networking (TON)*, Jul 2024.

[3] B. Hu, **Z. Huang**, N. Mehta, and N. Hegde, Near-Optimal Algorithms for Differentially Private Online Learning in a Stochastic Environment, *arXiv*, Jul 2024.

[4] K. Liu, J. Wang, **Z. Huang**, and J. Pan, Sampling-Based Multi-Job Placement for Heterogeneous Deep Learning Clusters, *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, Apr 2024.

[5] **Z. Huang** and J. Pan, Distributed Learning of Unknown Games for HetNet Selection, *IEEE Transactions on Network Science and Engineering (TNSE)*, Jan 2024.

[6] **Z. Huang**, Y. Xu, and J. Pan, TSOR: Thompson Sampling-based Opportunistic Routing, *IEEE Transactions on Wireless Communications (TWC)*, vol. 20, no. 11, pp. 7272-7285, May 2021.

[7] **Z. Huang**, B. Hu, and J. Pan, Caching by User Preference with Delayed Feedback for Heterogeneous Cellular Networks, *IEEE Transactions on Wireless Communications (TWC)*, vol. 20, no. 3, pp. 1655-1667, Mar 2021.

[8] R. Liu, J. Song, **Z. Huang**, and J. Pan, EQRC: A Secure QR Code-based E-coupon Framework Supporting Online and Offline Transactions, *Journal of Computer Security (JCS)*, vol. 28, no. 5, pp. 577-605, Sep 2020.

[9] **Z. Huang**, R. Zhang, J. Pan, Y. Jiang, and D. Zhai, A Framework of Multipath Clustering based on Space Transformed Fuzzy c-Means and Data Fusion for Radio Channel Modeling, *IEEE Transactions on Vehicular Technology (TVT)*, vol. 69, no. 1, pp. 4-15, Jan 2019.

TEACHING EXPERIENCE

Guest Lecturer, CSEP 546: Machine Learning, University of Washington — Autumn 2026

Instructor, CSC111: Fundamentals of Programming with Engineering Applications, University of Victoria — Spring 2024

Lab and Tutorial Instructor, CSC361: Computer Communications and Networks, University of Victoria — 2020 - 2023

Tutorial Instructor, CSC360: Operating Systems, University of Victoria — 2020 - 2023

PROFESSIONAL ACTIVITIES

PROGRAM COMMITTEE

- Shadow TPC member for ACM CoNEXT 2025

REVIEWER

- IEEE Transactions on Wireless Communications
- IEEE Transactions on Intelligent Transportation Systems
- IEEE Transactions on Artificial Intelligence
- IEEE Transactions on Signal Processing

AWARDS & SCHOLARSHIPS

Canada NSERC Postdoctoral Fellowship — 2025

IEEE INFOCOM 2025 Travel Award — 2025

British Columbia Graduate Scholarship — 2024

UVic President's Fellowship in Research-enriched Teaching — 2023

ICDCS 2023 Travel Award — 2023

UAI 2023 Scholarships — 2023

Melva J. Hanson Graduate Scholarship — 2022

IEEE ICDCS 2021 Best Poster Award — 2021

Howard E. Petch Research Scholarship — 2020